

**The due date for submitting this assignment has passed.**

**Due on 2026-03-25, 23:59 IST.**

**You may submit any number of times before the due date. The final submission will be considered for grading.**

**You have last submitted on: 2026-03-25, 07:24 IST**

**Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.**

**Multiple Select Questions (MSQ):**

1) A function  $T : V \rightarrow W$  between two vector spaces  $V$  and  $W$  is said to be a linear **1 point** transformation if the following conditions hold:

**Condition 1:**  $T(v_1 + v_2) = T(v_1) + T(v_2)$  for all  $v_1, v_2 \in V$ .

**Condition 2:**  $T(cv) = cT(v)$  for all  $v \in V$  and  $c \in \mathbb{R}$ .

Consider the following function:

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$T(x, y) = \begin{cases} 3x & \text{if } y = 0 \\ 4y & \text{if } y \neq 0 \end{cases}$$

Which of the following statements is true?

- ☐ Condition 1 holds.
- ☒ Condition 1 does not hold.
- ☒ Condition 2 holds.
- ☐ Condition 2 does not hold.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Condition 1 does not hold.

Condition 2 holds.

2) Suppose the matrix representation of a linear transformation  $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$  with **1 point** respect to ordered bases  $\beta = \{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$  for the domain and  $\gamma = \{(1, 0, 0), (0, 1, 0), (1, 0, 1)\}$  for the range, is  $I_{3 \times 3}$ , i.e., the identity matrix of order 3. Let  $A$  denote the matrix representation of the linear transformation  $T$  with respect to the standard ordered basis of  $\mathbb{R}^3$  for both domain and range. Which of the following are true?

- ☐  $A = I_{3 \times 3}$  i.e., identity matrix of order 3.
- ☐  $A$  is a singular matrix
- ☐

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$$

☒  $\det(A) = 1.$

☐  $\det(A) = -1.$

☒  $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$\det(A) = 1.$

$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

3) Match the linear transformations and sets of vectors in column A with the images of **1 point** those sets under the linear transformation in column B and the geometric representations of both sets in column C.

|      | Matrix form of linear transformation<br>(Column A)                                                                                                                                |    | Image of the given set<br>(Column B)      |    | Geometric representations<br>(Column C) |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------------------------------------|----|-----------------------------------------|
| i)   | $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ $T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 2 & 3 \end{bmatrix}$ <p>Set: <math>S = \{(x, y, z) \mid x + y + z = 1\}</math></p>  | a) | $T(S) = \{(x, y) \mid x - y = 1\}$        | 1) |                                         |
| ii)  | $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ $T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ <p>Set: <math>S = \{(x, y, z) \mid x + y + z = 1\}</math></p> | b) | $T(S) = \{(x, y, z) \mid x + y + z = 3\}$ | 2) |                                         |
| iii) | $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ $T = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ <p>Set: <math>S = \{(x, y) \mid x + y = 1\}</math></p>                             | c) | $T(S) = \{(x, y) \mid x - y = -1\}$       | 3) |                                         |
| iv)  | $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ $T = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ <p>Set: <math>S = \{(x, y) \mid x + y = 1\}</math></p>                              | d) | $T(S) = \{(x, y, z) \mid x + y - z = 1\}$ | 4) |                                         |

Choose the correct option from the following.

☐ i  $\rightarrow$  d  $\rightarrow$  2, ii  $\rightarrow$  b  $\rightarrow$  4.



i  $\rightarrow$  b  $\rightarrow$  4, ii  $\rightarrow$  d  $\rightarrow$  2.

☒ iii  $\rightarrow$  a  $\rightarrow$  1, iv  $\rightarrow$  c  $\rightarrow$  3

☐ iii  $\rightarrow$  a  $\rightarrow$  3, iv  $\rightarrow$  c  $\rightarrow$  1

No, the answer is incorrect.

Score: 0

Accepted Answers:

i  $\rightarrow$  b  $\rightarrow$  4, ii  $\rightarrow$  d  $\rightarrow$  2.

iii  $\rightarrow$  a  $\rightarrow$  3, iv  $\rightarrow$  c  $\rightarrow$  1

4) Consider a linear transformation  $S : M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$  such that  $S(A) = A^T$ . Let  $B$  **1 point** be the matrix representation of  $S$  with respect to the ordered bases:

$$\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

of  $M_2(\mathbb{R})$ . Choose the set of correct options:

☐ The order of the matrix  $B$  is  $2 \times 2$ .

☒ The order of the matrix  $B$  is  $4 \times 4$ .

☒ The dimension of the row space of the the matrix  $B$  is 4.

☐ The dimension of the column space of the matrix  $B$  is 3.

☐ The nullity of the the matrix  $B$  is 1.

☒ The rank of the the matrix  $B$  is 4.

☒  $S$  is surjective.

Yes, the answer is correct.

Score: 1

Accepted Answers:

The order of the matrix  $B$  is  $4 \times 4$ .

The dimension of the row space of the the matrix  $B$  is 4.

The rank of the the matrix  $B$  is 4.

$S$  is surjective.

5) Let  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  be a linear transformation and let  $\{v_1, v_2, \dots, v_n\}$  be a basis of  $\mathbb{R}^n$ . Suppose  $\{T(v_1), T(v_2), \dots, T(v_n)\}$  is a spanning set for  $\mathbb{R}^m$ . Choose all the correct statements. **1 point**

- ☒  $T$  is a surjective linear transformation.
- ☐  $T$  is a linear isomorphism.
- ☒  $m \leq n$ .
- ☒ For any basis  $\{w_1, w_2, \dots, w_n\}$  of  $\mathbb{R}^n$ , the set  $\{T(w_1), T(w_2), \dots, T(w_n)\}$  is a spanning set for  $\mathbb{R}^m$ .
- ☐ The matrix of the linear transformation  $T$  has rank equal to  $n$ .

Yes, the answer is correct.

Score: 1

Accepted Answers:

$T$  is a surjective linear transformation.

$m \leq n$ .

For any basis  $\{w_1, w_2, \dots, w_n\}$  of  $\mathbb{R}^n$ , the set  $\{T(w_1), T(w_2), \dots, T(w_n)\}$  is a spanning set for  $\mathbb{R}^m$ .

### Comprehension Type Question:

Suppose a bread-making machine B makes 6 breads from 2 eggs, 3 (in hundreds) grams of wheat, and 1 (in hundred) grams of sugar. B also makes 8 breads from 3 eggs, 4 (in hundreds) grams of wheat, and 2 (in hundreds) grams of sugar, and 10 breads from 5 eggs, 5 (in hundreds) grams of wheat, and 3 (in hundreds) grams of sugar. Suppose the production of breads is a linear function of the amount of eggs, wheat (in hundreds), and sugar (in hundreds) used as raw ingredients. Based on the above data answer the following questions. Suppose  $x$  eggs,  $y$  (in hundreds) grams of wheat, and  $z$  (in hundreds) grams of sugar are used as the raw materials to produce  $ax + by + cz$  number of breads. We can express this as follows:

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}$$

$$T(x, y, z) = ax + by + cz$$

where the co-ordinates in  $\mathbb{R}^3$  denote the number of eggs, amount (in grams) of wheat (in hundreds), and amount (in grams) of sugar (in hundreds). Observe that  $T$  is a linear transformation.

**Use the above information for Question 6, 7 and 8.**



6) Choose the correct set of options from the the following. (MSQ)

**1 point**

- ☐  $Nullity(T) = 1$
- ☒  $Rank(T) = 1$
- ☒  $Nullity(T) = 2$
- ☐  $Rank(T) = 2$
- ☐  $Nullity(T) = 3$
- ☐  $Rank(T) = 3$
- ☐  $T$  is neither one to one nor onto.
- ☐  $T$  is one to one but not onto.
- ☒  $T$  is onto but not one to one.
- ☐  $T$  is an isomorphism.

Yes, the answer is correct.

Score: 1

Accepted Answers:

$Rank(T) = 1$

$Nullity(T) = 2$

$T$  is onto but not one to one.

7) Choose the set of correct statements. (MSQ)

**1 point**

- ☐ If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then 9 breads are produced.
- ☒ If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then no bread is produced.
- ☒ If only 3 (in hundreds) grams of wheat is used, then 6 breads are produced.

- ☐ If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then no bread is produced.
- ☒ If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then 6 breads are produced.

Yes, the answer is correct.

Score: 1

Accepted Answers:

If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then no bread is produced.

If only 3 (in hundreds) grams of wheat is used, then 6 breads are produced.

If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then 6 breads are produced.

8) How many breads are produced by the machine from 6 eggs, 10 (in hundreds) grams of wheat, and 5 (in hundreds) grams of sugar? (NAT)

20

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 20

**1 point**

9) Let  $T$  be a linear transformation from  $\mathbb{R}^{10}$  to  $\mathbb{R}^{12}$ . Suppose a basis for the null space of  $T$  has 3 vectors. How many linearly independent vectors are needed to form a basis for the range of  $T$ ?

7

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 7

**1 point**

10) Let  $M$  be the set of all skew-symmetric matrices of order 7. Then  $M$  forms a vector space under matrix addition and scalar multiplication. Let  $T$  be a linear transformation from  $M$  to  $\mathbb{R}$  defined by  $T(A) = 6 \operatorname{tr}(A)$ , where  $\operatorname{tr}(A)$  = trace of  $A$ . What is the nullity of  $T$ ?

21

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 21.0

**1 point**

11) Let  $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$  be defined by  $T(x, y, z) = (10x + 9y, 2z)$ . Choose the correct **1 point** options about  $T$ .

- ☐ The matrix of  $T$  is  $\begin{bmatrix} 10 & 0 \\ 9 & 0 \\ 0 & 2 \end{bmatrix}$ .
- ☒ The matrix of  $T$  is  $\begin{bmatrix} 10 & 9 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ .
- ☒ A basis for the null space of  $T$  is  $\{(1, -\frac{10}{9}, 0)\}$ .
- ☐ A basis for the null space of  $T$  is  $\{(1, -\frac{10}{9}, 0), (0, 0, 2)\}$ .
- ☐ A basis for the range of  $T$  is  $\{(10, 10)\}$ .
- ☒ A basis for the range of  $T$  is  $\{(10, 9), (0, 2)\}$ .

Yes, the answer is correct.

Score: 1

Accepted Answers:

The matrix of  $T$  is  $\begin{bmatrix} 10 & 9 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ .

A basis for the null space of  $T$  is  $\{(1, -\frac{10}{9}, 0)\}$ .

A basis for the range of  $T$  is  $\{(10, 9), (0, 2)\}$ .